

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

## LECTURE 10.2

# Solid Oxide Cells & Optimisation

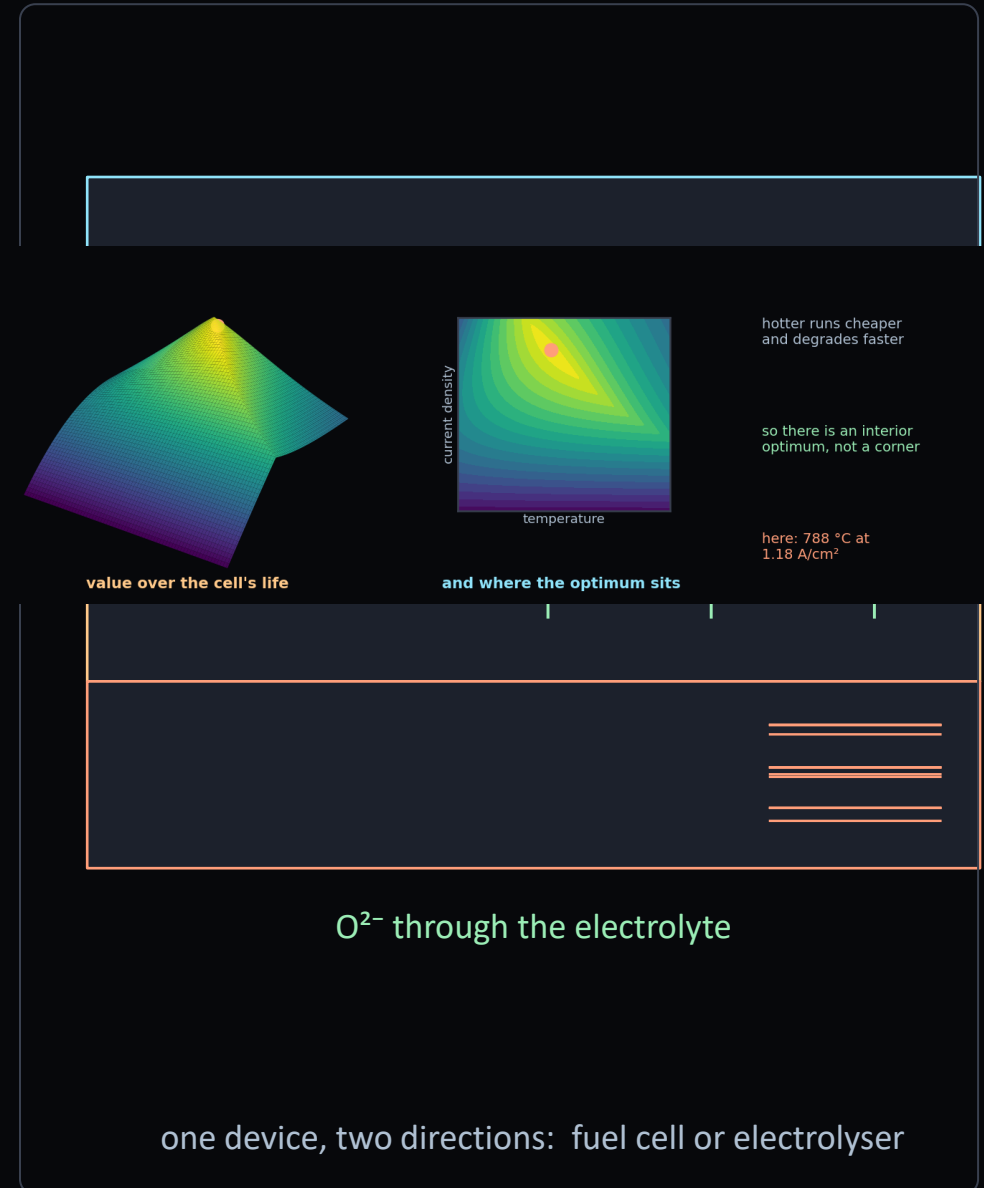
The same three physics at 800 °C — and the point the whole course has been building towards: a differentiable model inside a lifetime-aware optimiser

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*Public SOFC/SOEC literature · Gu & Wang (2000) for the thermal method*



# What You Will Be Able To Do

## THE GOALS, IN PLAIN WORDS

- **read** — a polarisation curve, and name which loss dominates where
- **explain** — the thermoneutral voltage, and why stacks are designed around it
- **state** — why a hot spot in a solid oxide cell is self-reinforcing
- **write** — an objective with a degradation constraint, not just a performance one
- **argue** — why optimisation needs a differentiable model, not just an accurate one
- **recognise** — where the course has been heading since L7

## WHAT TODAY ASSUMES

|            |                                        |
|------------|----------------------------------------|
| from L10.1 | coupled electrochemistry and heat      |
| from L9    | convection-diffusion in a channel      |
| from L8    | thermal gradients, and why they matter |

## THE LAST PHYSICS LECTURE

Four of the six rows in the physics table are already familiar. What is new is that the answer is an operating schedule, not a field.

## AND A DIFFERENT KIND OF QUESTION

Not what is the best operating point, but what is the best trajectory.



# Four of Six Rows Are Already Familiar

| PHYSICS             | IN A SOLID OXIDE CELL             | WHERE YOU MET IT       |
|---------------------|-----------------------------------|------------------------|
| Species transport   | convection–diffusion in a channel | L9 — same operator     |
| Charge conservation | elliptic, ionic conduction        | L10.1 — same operator  |
| Heat with a source  | parabolic, electrochemical source | L8.2, L10.1            |
| Reaction kinetics   | Butler–Volmer at the interface    | L10.1 — same form      |
| Temperature         | 600–900 °C                        | new — and it dominates |
| Reversibility       | same device runs both ways        | new                    |

Almost nothing here is new physics. What is new is the operating temperature, which turns thermal gradients from a performance issue into the thing that decides how long the device lives.

# What Gets Optimised, and With What

## WHAT YOU ALREADY HAVE

- the physics network is L10.1's, with a channel coordinate instead of a particle radius
- what is new is that the network is inside an optimisation, not the end of one
- L6.1's autograd supplies the gradient of the objective, not only of the loss
- and that is the whole argument for a differentiable model

## THE NETWORK, CONCRETELY

|                        |                                          |
|------------------------|------------------------------------------|
| <b>physics network</b> | channel position and time in, fields out |
| <b>from L10.1</b>      | the same coupled-field structure         |
| <b>from L6.1</b>       | autograd, now on the objective           |
| <b>new here</b>        | a decision variable, not just a field    |

## THE GRADIENT IS THE POINT

A classical solver evaluates the physics. Autograd differentiates through it, so the schedule is optimised directly.

## AND L11.2 DOES THE SAME THING

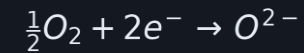
A robot car instead of a stack, same argument.

# One Device, Two Directions

## FUEL ELECTRODE (SOFC MODE)



## OXYGEN ELECTRODE (SOFC MODE)



- oxygen ions conduct through a dense ceramic electrolyte
- **run it one way and hydrogen and oxygen make electricity** — a fuel cell
- **reverse the current and it splits steam into hydrogen** — an electrolyser
- the fuel electrode is the anode in fuel-cell mode and the cathode in electrolysis

For a PINN this is an unusually good property: a single trained model covers both operating modes, because the sign of the current is an input rather than a different physics.

# The Polarisation Curve

## NERNST POTENTIAL — THE REVERSIBLE VOLTAGE

$$E = E^0(T) + \frac{RT}{2F} \ln \left( \frac{p_{H_2} p_{O_2}^{1/2}}{p_{H_2O}} \right)$$

$E^0$  the standard potential,  $p$  the partial pressures  
 $R$  the gas constant,  $F$  Faraday's constant  
 $\eta_{act}, \eta_{ohm}, \eta_{conc}$  the three losses

## FUEL CELL

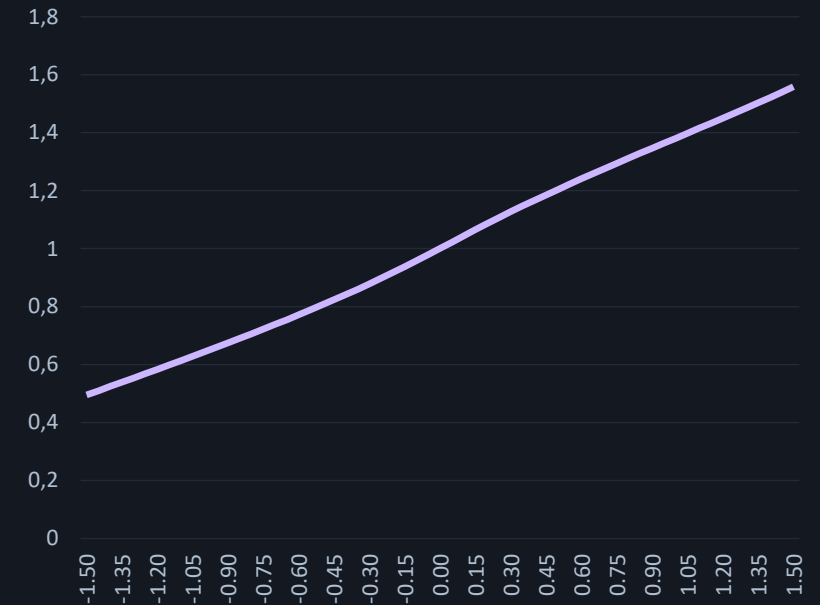
$$V = E - \eta_{act} - \eta_{ohm} - \eta_{conc}$$

## ELECTROLYSER

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

- three losses, each dominant in a different current range
- activation kinetics dominate at low current and concentration losses at high
- **note the sign flip** — in fuel-cell mode the overpotentials subtract
- in electrolysis they add
- ohmic loss is current times resistance, so it always rises with current

## I-V IN BOTH DIRECTIONS



Current density on the horizontal axis. Positive is electrolysis, negative is fuel-cell mode; the curve passes smoothly through the open-circuit voltage at zero.

# The Thermoneutral Voltage

## WHERE ELECTRICAL INPUT EXACTLY MATCHES THE REACTION ENTHALPY

$$V_{tn} = \frac{\Delta H}{2F} \approx 1.29 \text{ V} \quad (\text{steam, } 800^\circ\text{C})$$

$\Delta H$  the reaction enthalpy,  $F$  Faraday's constant  
 $q$  the net heat,  $i$  the current density

## NET HEAT GENERATION

$$q = i(V - V_{tn})$$

- below the thermoneutral voltage the cell absorbs heat
- above it, the cell generates heat and must be cooled
- most electrolyser stacks are designed around that point
- the heat source has an unusually elegant form
- one expression replaces the four-term source of L10.1

## THREE OPERATING REGIMES

### endothermal

$$V < V_{tn}$$

heat must be supplied; efficiency above 100% electrical; lowest degradation

### thermoneutral

$$V = V_{tn}$$

net heat flux zero; the design point for most stacks

### exothermal

$$V > V_{tn}$$

cell must be cooled; higher production, faster ageing



# Gas Transport Along the Channel

## SPECIES TRANSPORT ALONG THE FLOW DIRECTION

$$\frac{\partial c_k}{\partial t} + u \frac{\partial c_k}{\partial x} = D_k \frac{\partial^2 c_k}{\partial x^2} + S_k(i)$$

$c_k$  the concentration of species  $k$ ,  $u$  the gas speed,  $D_k$  its diffusivity  
 $S_k(i)$  the reaction source,  $A$  the cell area,  $\dot{n}_{H_2}$  the hydrogen produced

## REACTION SOURCE

$$\dot{n}_{H_2} = \frac{iA}{2F}$$

- convection dominates in the gas channel; diffusion matters in the porous layers
- composition changes along the channel, so the Nernst potential does too
- the applied voltage is essentially uniform; it is the current density that varies
- which is why a zero-dimensional model is a starting point, not an answer
- reactant utilisation is the design variable

# The Coupled System

## CHARGE CONSERVATION — ELLIPTIC

$$\nabla \cdot (\sigma_{ion} \nabla \phi) = 0, \quad i = -\sigma_{ion} \nabla \phi$$

## RESISTANCE FALLS SHARPLY WITH TEMPERATURE

$$ASR(T) = ASR_0 \exp \left[ \frac{E_a}{R} \left( \frac{1}{T} - \frac{1}{T_0} \right) \right]$$

## ENERGY CONSERVATION WITH THE ELECTROCHEMICAL SOURCE

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) + i(V - V_{tn})$$

$\sigma_{ion}$  the ionic conductivity,  $\phi$  the potential,  $i$  the current density  
 $ASR$  the area specific resistance,  $E_a$  its activation energy,  $V_{tn}$  thermoneutral

- two heat sources: ohmic, from resistance to current, and electrochemical, from the reactions
- and the coupling is stronger here than in a battery
- the resistance is Arrhenius in temperature, so a hot spot draws more current
- which makes it hotter
- in a battery a gradient degrades performance; here it cracks ceramic

# Start With the Simplest Model

## BUTTON CELL

### 0-D

Uniform everything.  
One algebraic voltage balance  
plus a lumped temperature.  
Fits published polarisation data.  
Where the exercise starts.

## CHANNEL

### 1-D

Composition varies along the  
flow; current density with it.  
Adds the L9 transport operator.  
Captures utilisation effects.  
The realistic teaching model.

## REPEAT UNIT

### 2-D / stack

Cross-flow or counter-flow,  
with in-plane thermal gradients.  
This is where thermal stress  
and lifetime actually live.  
A stretch target.

- **the button-cell model is not a toy** — zero-dimensional is a modelling choice, not a size
- so it is what you can actually compare against; fit it to a published curve first
- only then add the channel, where the gradients appear

# What to Check, and Against What

## POLARISATION CURVE

The primary check. Published  $i$ - $V$  curves exist for button cells at stated temperature and gas composition — reproduce one before believing anything else.

## OPEN-CIRCUIT VOLTAGE

At zero current the model must return the Nernst potential for the stated composition. A pure thermodynamic check, with no kinetics involved.

## THERMONEUTRAL POINT

The voltage at which net heat generation crosses zero should come out near 1.29 V for steam at high temperature. If not, the enthalpy or the sign convention is wrong.

## TEMPERATURE SENSITIVITY

Repeat the polarisation curve at two temperatures and check that the ohmic slope changes by the Arrhenius factor. This tests the coupling, not just the steady state.

Unlike the battery half of L10, there is no PyBaMM here — no single community reference implementation. That makes these four checks the whole of your verification, and it makes stating your data source non-negotiable.

# What Actually Kills a Solid Oxide Cell

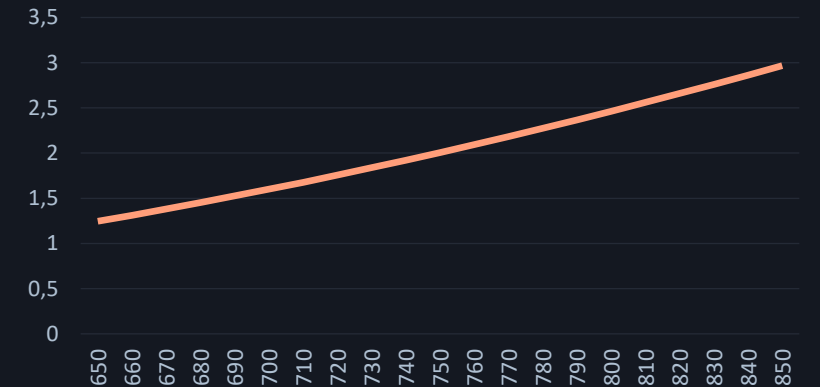
## A USABLE EMPIRICAL DEGRADATION LAW

$$\frac{d ASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^m$$

$ASR$  the area specific resistance,  $i$  the current density  
 $E_d$  the activation energy of degradation,  $m$  an exponent

- degradation shows up as a rising area-specific resistance
- **the mechanisms are microstructural** — nickel coarsening, delamination, chromium poisoning
- the published numbers span an enormous range
- the law below is a teaching assumption; real degradation is rarely linear in time
- many factors drive it, so treat any single expression as indicative only

## DEGRADATION GROWS WITH T



*Illustrative Arrhenius shape against temperature in °C. Higher temperature generally accelerates degradation, though many other factors contribute.*



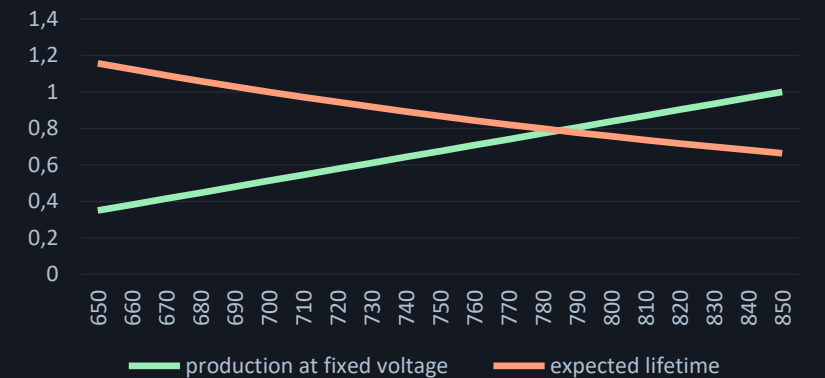
# The Trade-Off That Makes This an Optimisation Problem

**BOTH MOVE THE SAME WAY WITH TEMPERATURE — AND THAT IS THE WHOLE PROBLEM**

$\dot{n}_{H_2} \uparrow$  with  $T$ ,  $\frac{dASR}{dt} \uparrow$  with  $T$       $\dot{n}_{H_2}$  the hydrogen produced,  $ASR$  the resistance,  $T$  the operating temperature

- raise the temperature and resistance falls, so the cell consumes less energy now
- but degradation generally accelerates, so it produces less later
- reducing operating temperature is a recognised strategy for slowing that
- the trade-off is better stated in current density: higher density, more output, more degradation
- so the question is not the best operating point but the best trajectory

## PULLING IN OPPOSITE DIRECTIONS



operating temperature → (schematic, at fixed voltage)

# Writing Down What You Actually Want

## VALUE PRODUCED OVER THE LIFE OF THE DEVICE

$$\max_{i(t), T(t)} \int_0^{t_{life}} [p_{H_2} \dot{n}_{H_2} - c_e i A V] dt$$

$p_{H_2}$  the hydrogen price,  $c_e$  the electricity price,  $A$  the cell area,  $t_{life}$  the design life

## SUBJECT TO END-OF-LIFE AND THERMAL-STRESS LIMITS

$$\text{subject to } ASR(t_{life}) \leq ASR_{max}, \quad |\nabla T| \leq |\nabla T|_{max}$$

- **two terms in the objective** — hydrogen is worth something, electricity costs something
- two constraints that matter
- the resistance must not exceed its end-of-life threshold before the design lifetime
- and the thermal gradient must stay below what the ceramic tolerates
- the second is what makes a distributed model necessary

# Why a Differentiable Model Changes the Problem

## RECEDING-HORIZON CONTROL WITH A LEARNED MODEL

$$\min_{u_{0..N}} \sum_{k=0}^N \ell(x_k, u_k) \quad \text{with } x_{k+1} = \mathcal{N}_{\hat{w}}(x_k, u_k)$$

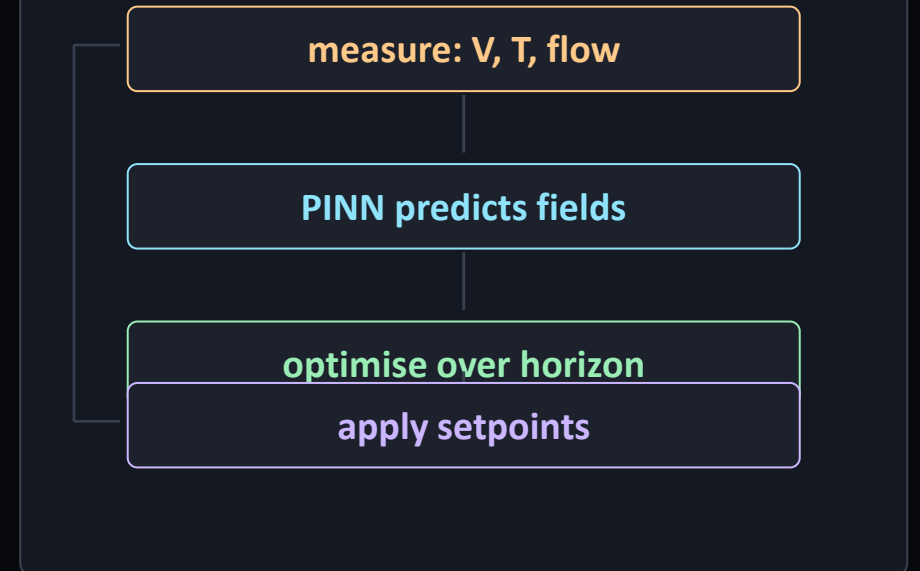
## AND THE GRADIENT IS FREE

$$\frac{\partial J}{\partial u} \text{ from the same autograd}$$

- a classical solver evaluates; to optimise you finite-difference or adjoint
- a PINN is differentiable by construction
- so the gradient with respect to the operating schedule is available directly
- that is the difference between optimising offline and optimising in the loop
- and it makes a constraint on the thermal gradient tractable

$x_k$  the state,  $u_k$  the control at step  $k$   
 $\ell$  the cost of one step,  $J$  their sum  
 $\mathcal{N}_{\hat{w}}$  the network, as the model

## THE CONTROL LOOP



# What Each Predicted Field Buys You

| PREDICTED FIELD       | FROM      | WHAT IT DECIDES                                         |
|-----------------------|-----------|---------------------------------------------------------|
| Temperature field     | L8, L10.1 | the degradation rate, and the thermal-stress constraint |
| Thermal gradient      | L8.1      | the crack criterion — a lumped model cannot express it  |
| Species concentration | L9, L10.2 | local Nernst potential and the utilisation limit        |
| Flow field            | L9.1      | how heat and reactant are actually distributed          |
| Resistance $ASR(t)$   | L10.2     | state of health, and how far from end of life           |

Read the middle column. Every lecture in this course produces one row of this table, and the optimisation is what consumes them all at once.

# Where to Be Careful

## Degradation laws are empirical

The Arrhenius-times-power form is a fit, not a mechanism, and published rates vary by orders of magnitude.

Never present a predicted lifetime as a number. Present a trend, and state the fitted law.

## Extrapolating in time

A model fitted to 1000 hours of data used to predict 40 000 hours of operation.

The dominant mechanism can change with age — early degradation often differs from steady behaviour.

## Optimising a wrong model

A gradient-based optimiser will exploit any error in the surrogate, confidently.

Constrain the optimiser to the region where the model was validated, and say where that is.

## No community reference

Unlike PyBaMM for batteries, there is no single trusted open implementation to check against.

Cite the exact source of every parameter and every data set you fit to.

## Thermal stress is structural

Cracking depends on mechanical properties this model does not contain.

A gradient limit is a proxy. Treat it as a screening criterion, not a failure prediction.

*The third is the one that bites in practice: an optimiser is an adversary that finds your model's weakest point and operates there.*

# Questions

## 1 – 2 · SOLID OXIDE CELL

### 1 What is a solid oxide cell, and what does its polarisation curve show?

One device that runs as a fuel cell or, with the current reversed, as an electrolyser, at about 800 °C. The curve is voltage against current density: the activation loss dominates at low current, the ohmic loss in the middle, and gas transport at high current. *Ex\_10.2 notebooks 00 – 01*

### 2 What is the thermoneutral voltage, and why is a stack designed around it?

In electrolysis, the voltage at which the heat the reaction absorbs equals the heat the losses release: the net heat is zero and the temperature stays even. Above it the cell produces more, and pays with heat, thermal gradients and life. *Ex\_10.2 notebooks 01 – 02*

## 3 – 4 · LIFETIME OPTIMISATION

### 3 Why is operating a cell an optimisation problem and not a setting?

A higher temperature raises both the production and the degradation rate, so there is no best operating point, only a best trajectory. It follows the hourly electricity price and is held back by a limit on the lifetime it may use. *Ex\_10.2 notebooks 03 – 04*

### 4 Why does optimisation want a differentiable model, and what should you refuse to claim from it?

A classical solver evaluates the physics; a network also gives the gradient of the objective with respect to the operating schedule, which is what an optimiser needs. Degradation rates are uncertain by an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5; Ex\_10.2 notebook 04*

## Ex\_10.2 — From Button Cell to Lifetime Optimisation

### TASKS

- 1 Fit a 0-D button-cell model to a synthetic polarisation curve and verify the open-circuit voltage.
- 2 Locate the thermoneutral point and confirm that net heat generation crosses zero there.
- 3 Extend to a 1-D channel and show how utilisation shifts the local Nernst potential.
- 4 Add the degradation law and simulate resistance growth at three operating temperatures.
- 5 Optimise the operating trajectory against a time-varying electricity price, subject to a lifetime constraint.

### QUESTIONS, AND WHERE ANSWERED

Why is the thermoneutral point special? sl. 7

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Why does a lumped model fail for lifetime? sl. 14

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What makes temperature a trade-off rather than a setting? sl. 13

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Why does differentiability change what is possible? sl. 15

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What would you refuse to claim from your result? sl. 17

*The last question carries as many marks as the first four. Knowing what your model cannot support is the skill this course is really teaching.*



# Key Takeaways

- 1 Same physics, hotter**  
Species transport, charge conservation and heat — the operators of L9 and L10.1. What changes is 800 °C, and thermal gradients end the device.
- 2 One device, two directions**  
Fuel cell and electrolyser are the same hardware with the current reversed. One model serves both: the current's sign is an input, not new physics.
- 3 Thermoneutral is the design point**  
At  $V_{tn}$  the net heat flux is zero and stack thermal management becomes trivial. Everything above it is production bought with heat and with life.
- 4 Lifetime is the real objective**  
Temperature raises both production and degradation. There is no best setting, only a best trajectory, and it follows hourly prices.
- 5 Differentiability is the point**  
A classical solver evaluates the physics; a PINN differentiates it. That is what moves optimisation from offline scenarios to online control.

*Next — L11: robotics, where the physics-informed idea survives without a governing PDE at all.*

# Sources and Where to Look Next

## Thermoneutral operation

SOEC stacks are typically operated at the thermoneutral voltage for steam splitting, about 1.29 V, against roughly 1.48 V for low-temperature water electrolysis — a stack-level efficiency gain of order 15%.

## Temperature and degradation

Long-term SOEC tests consistently report faster degradation at higher operating temperature. Lowering temperature is a recognised mitigation, at the cost of output or of electricity — the trade-off this lecture is built on.

## Long-duration benchmarks

Multi-thousand-hour cell and stack tests exist in the open literature, with degradation traced by impedance spectroscopy. These are the data sets to fit against.

## Gu & Wang (2000)

The thermal-electrochemical coupling method of L10.1 transfers directly: the heat source, the Arrhenius property dependence, and the argument against decoupled models.

## Find the current numbers yourself

Degradation rates and cell performance move quickly. Locating and citing the state of the art is part of the exercise — as it was for turbulence in L9.2.

*No PyBaMM equivalent exists for solid oxide cells, so every parameter and data set must be cited explicitly.*